

Day - 6

PROMETHEUS BASICS

Prometheus is an open-source system monitoring and alerting toolkit originally built at SoundCloud. It is now a standalone opensource project. Prometheus joined the Cloud Native Computing Foundation in 2016 as the second hosted project, after Kubernetes.

Features

1. A multi-dimensional data model with time series data identified by metric name and key/value pairs
2. PromQL, a flexible query language to leverage this dimensionality
3. No reliance on distributed storage; single server nodes are autonomous
4. Time series collection happens via a pull model over HTTP
5. Pushing time series is supported via an intermediary gateway
6. Targets are discovered via service discovery or static configuration
7. Multiple modes of graphing and dashboarding support

Prometheus Installation: (Commands with output)

```
:
sudo useradd \
  --system \
  --no-create-home \
  --shell /bin/false prometheus
```

```
wget
https://github.com/prometheus/prometheus/releases/download/v2.
47.1/prometheus-2.47.1.linux-amd64.tar.gz

tar -xvf prometheus-2.47.1.linux-amd64.tar.gz

sudo mkdir -p /data /etc/prometheus

cd prometheus-2.47.1.linux-amd64/

sudo mv prometheus promtool /usr/local/bin/

sudo mv consoles/ console_libraries/ /etc/prometheus/

sudo mv prometheus.yml /etc/prometheus/prometheus.yml

sudo chown -R prometheus:prometheus
/etc/prometheus/ /data/
```

```
cd

rm -rf prometheus-2.47.1.linux-amd64.tar.gz

prometheus --version

sudo vim /etc/systemd/system/prometheus.service
```

```
Ubuntu 22.04.5 LTS  X + v
ubuntu prometheus-2.47.1.linux-amd64 12:02 cd prometheus-2.47.1.linux-amd64/
console_libraries/ prometheus-2.47.1.linux-amd64/
consoles/
ubuntu prometheus-2.47.1.linux-amd64 12:02 cd prometheus-2.47.1.linux-amd64/
ubuntu prometheus-2.47.1.linux-amd64 12:02 ls
LICENSE console_libraries prometheus-2.47.1.linux-amd64
NOTICE consoles prometheus-2.47.1.linux-amd64.tar.gz
ubuntu prometheus-2.47.1.linux-amd64 12:02 rm -rf prometheus-2.47.1.linux-amd64.tar.gz
ubuntu prometheus-2.47.1.linux-amd64 12:03 prometheus --version
prometheus, version 2.47.1 (branch: HEAD, revision: c4d1a8beff37cc004f1dc4ab9d2e73193f51aaeb)
  build user: root@4829330363be
  build date: 20231004-10:31:16
  go version: go1.21.1
  platform: linux/amd64
  tags: netgo,builtinassets,stringlabels
ubuntu prometheus-2.47.1.linux-amd64 12:03 sudo vim /etc/systemd/system/prometheus.service
ubuntu prometheus-2.47.1.linux-amd64 12:03 ls
LICENSE NOTICE console_libraries consoles prometheus-2.47.1.linux-amd64
ubuntu prometheus-2.47.1.linux-amd64 12:03 cd prometheus-2.47.1.linux-amd64/
ubuntu prometheus-2.47.1.linux-amd64 12:04 ls
LICENSE NOTICE console_libraries consoles
ubuntu prometheus-2.47.1.linux-amd64 12:04 ls
LICENSE NOTICE console_libraries consoles
ubuntu prometheus-2.47.1.linux-amd64 12:05 prometheus --version
Command 'prometheus' not found, did you mean:
  command 'prometheus' from deb prometheus (2.31.2+ds1-lubuntu1.22.04.3)
Try: sudo apt install <deb name>
ubuntu prometheus-2.47.1.linux-amd64 12:05 prometheus --version
prometheus, version 2.47.1 (branch: HEAD, revision: c4d1a8beff37cc004f1dc4ab9d2e73193f51aaeb)
  build user: root@4829330363be
  build date: 20231004-10:31:16
  go version: go1.21.1
  platform: linux/amd64
  tags: netgo,builtinassets,stringlabels
ubuntu prometheus-2.47.1.linux-amd64 12:05 sudo vim /etc/systemd/system/prometheus.service
ubuntu prometheus-2.47.1.linux-amd64 12:06
```

[Unit]

Description=Prometheus

Wants=network-online.target

After=network-online.target

StartLimitIntervalSec=500

StartLimitBurst=5

[Service]

User=prometheus

Group=prometheus

Type=simple

Restart=on-failure

RestartSec=5s

ExecStart=/usr/local/bin/prometheus \

--config.file=/etc/prometheus/prometheus.yml \

--storage.tsdb.path=/data \

--web.console.templates=/etc/prometheus/consoles \

--web.console.libraries=/etc/prometheus/console_libraries \

--web.listen-address=0.0.0.0:9090 \

--web.enable-lifecycle

[Install]

WantedBy=multi-user.target

sudo systemctl enable prometheus

sudo systemctl start prometheus

sudo systemctl status prometheus

journalctl -u prometheus -f --no-pager

```
Ubuntu 22.04.5 LTS
● prometheus.service - Prometheus
   Loaded: loaded (/etc/systemd/system/prometheus.service; enabled; vendor preset: enabled)
   Active: active (running) since Sat 2025-03-22 12:10:15 IST; 6s ago
     Main PID: 46336 (prometheus)
        Tasks: 15 (Limit: 9299)
       Memory: 26.3M
      CGroup: /system.slice/prometheus.service
              └─46336 /usr/local/bin/prometheus --config.file=/etc/prometheus/prometheus.yml --storage.tsdb.path=/data --web.console.templates=/etc/prometheus/consoles --web.external-url=https://localhost:9090

Mar 22 12:10:15 NIDHARSAN prometheus[46336]: ts=2025-03-22T06:40:15.810Z caller=head.go:681 level=info component=tsdb msg="On-disk memory index loaded"
Mar 22 12:10:15 NIDHARSAN prometheus[46336]: ts=2025-03-22T06:40:15.810Z caller=head.go:689 level=info component=tsdb msg="Replaying WAL segment"
Mar 22 12:10:15 NIDHARSAN prometheus[46336]: ts=2025-03-22T06:40:15.813Z caller=head.go:760 level=info component=tsdb msg="WAL segment"
Mar 22 12:10:15 NIDHARSAN prometheus[46336]: ts=2025-03-22T06:40:15.813Z caller=head.go:797 level=info component=tsdb msg="WAL replay completed"
Mar 22 12:10:15 NIDHARSAN prometheus[46336]: ts=2025-03-22T06:40:15.814Z caller=main.go:1045 level=info fs_type=EXT4_SUPER_MAGIC
Mar 22 12:10:15 NIDHARSAN prometheus[46336]: ts=2025-03-22T06:40:15.814Z caller=main.go:1048 level=info msg="TSDB started"
Mar 22 12:10:15 NIDHARSAN prometheus[46336]: ts=2025-03-22T06:40:15.814Z caller=main.go:1229 level=info msg="Loading configuration file" filename=/etc/prometheus/prometheus.yml
Mar 22 12:10:15 NIDHARSAN prometheus[46336]: ts=2025-03-22T06:40:15.815Z caller=main.go:1266 level=info msg="Completed loading of configuration file" filename=/etc/prometheus/prometheus.yml
Mar 22 12:10:15 NIDHARSAN prometheus[46336]: ts=2025-03-22T06:40:15.815Z caller=main.go:1009 level=info msg="Server is ready to receive web requests."
Mar 22 12:10:15 NIDHARSAN prometheus[46336]: ts=2025-03-22T06:40:15.815Z caller=manager.go:1009 level=info component="rule manager" msg="Starting rule manager"

lines 1-19/19 (END)
```

```
Ubuntu 22.04.5 LTS
journalctl -u prometheus -f --no-pager
Mar 22 12:10:15 NIDHARSAN prometheus[46336]: ts=2025-03-22T06:40:15.810Z caller=head.go:681 level=info component=tsdb msg="On-disk memc
play completed" duration=2.325µs
Mar 22 12:10:15 NIDHARSAN prometheus[46336]: ts=2025-03-22T06:40:15.810Z caller=head.go:689 level=info component=tsdb msg="Replaying WA
ile"
Mar 22 12:10:15 NIDHARSAN prometheus[46336]: ts=2025-03-22T06:40:15.813Z caller=head.go:760 level=info component=tsdb msg="WAL segment loaded" segment=0 max
Segment=0
Mar 22 12:10:15 NIDHARSAN prometheus[46336]: ts=2025-03-22T06:40:15.813Z caller=head.go:797 level=info component=tsdb msg="WAL replay completed" checkpoint_
replay_duration=12.966µs wal_replay_duration=3.188997ms wbl_replay_duration=367ns total_replay_duration=3.221384ms
Mar 22 12:10:15 NIDHARSAN prometheus[46336]: ts=2025-03-22T06:40:15.814Z caller=main.go:1045 level=info fs_type=EXT4_SUPER_MAGIC
Mar 22 12:10:15 NIDHARSAN prometheus[46336]: ts=2025-03-22T06:40:15.814Z caller=main.go:1048 level=info msg="TSDB started"
Mar 22 12:10:15 NIDHARSAN prometheus[46336]: ts=2025-03-22T06:40:15.814Z caller=main.go:1229 level=info msg="Loading configuration file" filename=/etc/promet
heus/prometheus.yml
Mar 22 12:10:15 NIDHARSAN prometheus[46336]: ts=2025-03-22T06:40:15.815Z caller=main.go:1266 level=info msg="Completed loading of configuration file" filena
me=/etc/prometheus/prometheus.yml totalDuration=392.396µs db_storage=744ns remote_storage=1.712µs web_handler=391ns query_engine=652ns scrape=152.903µs scra
pe_sd=26.798µs notify=21.445µs notify_sd=5.027µs rules=1.147µs tracing=11.986µs
Mar 22 12:10:15 NIDHARSAN prometheus[46336]: ts=2025-03-22T06:40:15.815Z caller=main.go:1009 level=info msg="Server is ready to receive web requests."
Mar 22 12:10:15 NIDHARSAN prometheus[46336]: ts=2025-03-22T06:40:15.815Z caller=manager.go:1009 level=info component="rule manager" msg="Starting rule manag
er..."
```

Node Exporter:

```
sudo useradd \
    --system \
    --no-create-home \
    --shell /bin/false node_exporter
```

wget

https://github.com/prometheus/node_exporter/releases/download/v1.6.1/node_exporter-1.6.1.linux-amd64.tar.gz

tar -xvf node_exporter-1.6.1.linux-amd64.tar.gz

```
Ubuntu 22.04.5 LTS
C-SHA256GX-Amz-Credential=releaseassetproduction%2F20250322%2Fus-east-1%2Fs3%2Faws4_request&X-Amz-Date=20250322T064212Z&X-Amz-Expires=3
4b419e506787341bcb340d11ea6b0793b94579632191ef86259c8bbdc3f51&X-Amz-SignedHeaders=host&response-content-disposition=attachment%3B%20fi
r-1.6.1.linux-amd64.tar.gz&response-content-type=application%2Foctet-stream [following]
--2025-03-22 12:12:30-- https://objects.githubusercontent.com/github-production-release-asset-2e65be/9524057/5509b569-5c34-471e-8598-c
orithm=AWS4-HMAC-SHA256GX-Amz-Credential=releaseassetproduction%2F20250322%2Fus-east-1%2Fs3%2Faws4_request&X-Amz-Date=20250322T064212Z&
mz-Signature=984b419e506787341bcb340d11ea6b0793b94579632191ef86259c8bbdc3f51&X-Amz-SignedHeaders=host&response-content-disposition=att
%3Dnode_exporter-1.6.1.linux-amd64.tar.gz&response-content-type=application%2Foctet-stream
Resolving objects.githubusercontent.com (objects.githubusercontent.com)...1185.199.108.133, 185.199.109.133, 185.199.111.133, ...
Connecting to objects.githubusercontent.com (objects.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 10368103 (9.9M) [application/octet-stream]
Saving to: 'node_exporter-1.6.1.linux-amd64.tar.gz'

node_exporter-1.6. 100%[=====] 9.89M 9.87MB/s in 1.0s

2025-03-22 12:12:33 (9.87 MB/s) - 'node_exporter-1.6.1.linux-amd64.tar.gz' saved [10368103/10368103]

node_exporter-1.6.1.linux-amd64/
node_exporter-1.6.1.linux-amd64/NOTICE
node_exporter-1.6.1.linux-amd64/node_exporter
node_exporter-1.6.1.linux-amd64/LICENSE
ubuntu@prometheus-2.47.1.linux-amd64:~$ sudo mv \
node_exporter-1.6.1.linux-amd64/node_exporter \
/usr/local/bin/
ubuntu@prometheus-2.47.1.linux-amd64:~$ rm -rf node_exporter*
ubuntu@prometheus-2.47.1.linux-amd64:~$ node_exporter --version
node_exporter, version 1.6.1 (branch: HEAD, revision: 4a1b7760c1873a8233f3fffb55afcedbb63b8d84)
 build user: root@586879db11e5
 build date: 20230717-12:10:52
 go version: go1.20.6
 platform: linux/amd64
 tags: netgo osusergo static_build
ubuntu@prometheus-2.47.1.linux-amd64:~$
```


`promtool check config /etc/prometheus/prometheus.yml`



```
prometheus-2.47.1-linux-amd64 promtool check config /etc/prometheus/prometheus.yml
Checking /etc/prometheus/prometheus.yml
SUCCESS: /etc/prometheus/prometheus.yml is valid prometheus config file syntax
```

Grafana :

`sudo apt-get install -y apt-transport-https software-properties-common`

`wget -q -O - https://packages.grafana.com/gpg.key | sudo apt-key add -`

`echo "deb https://packages.grafana.com/oss/deb stable main" |
sudo tee -a /etc/apt/sources.list.d/grafana.list`

`sudo apt-get update`

`sudo apt-get -y install grafana`

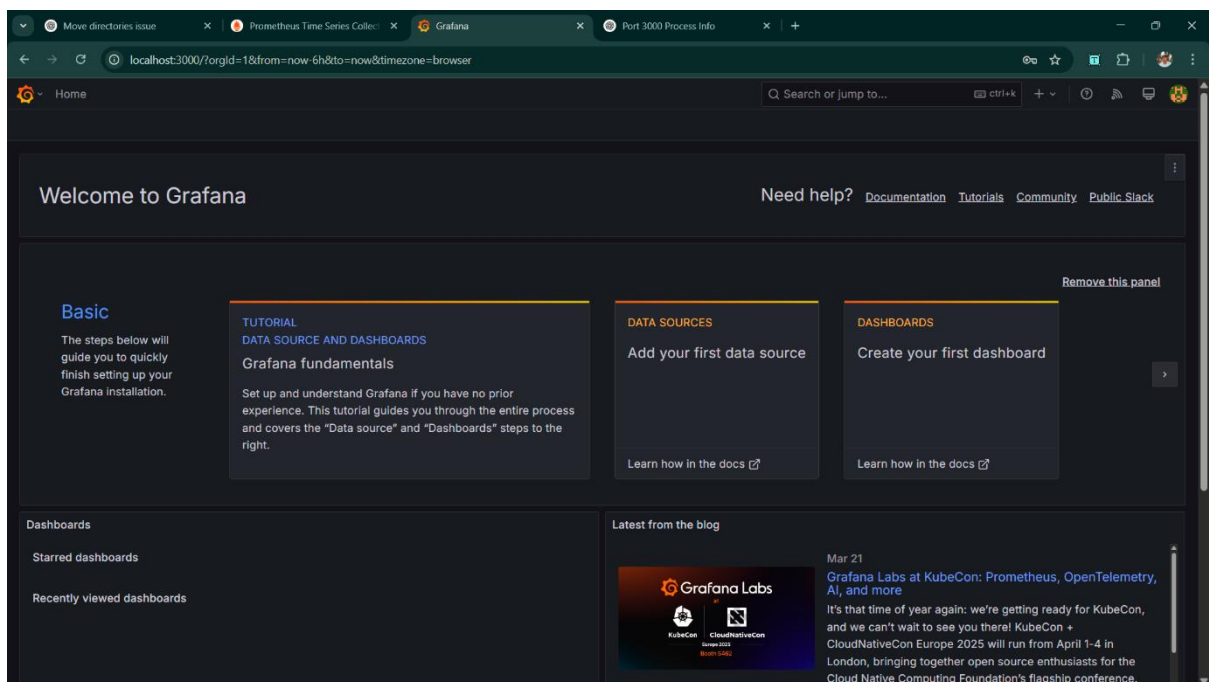
`sudo systemctl enable grafana-server`

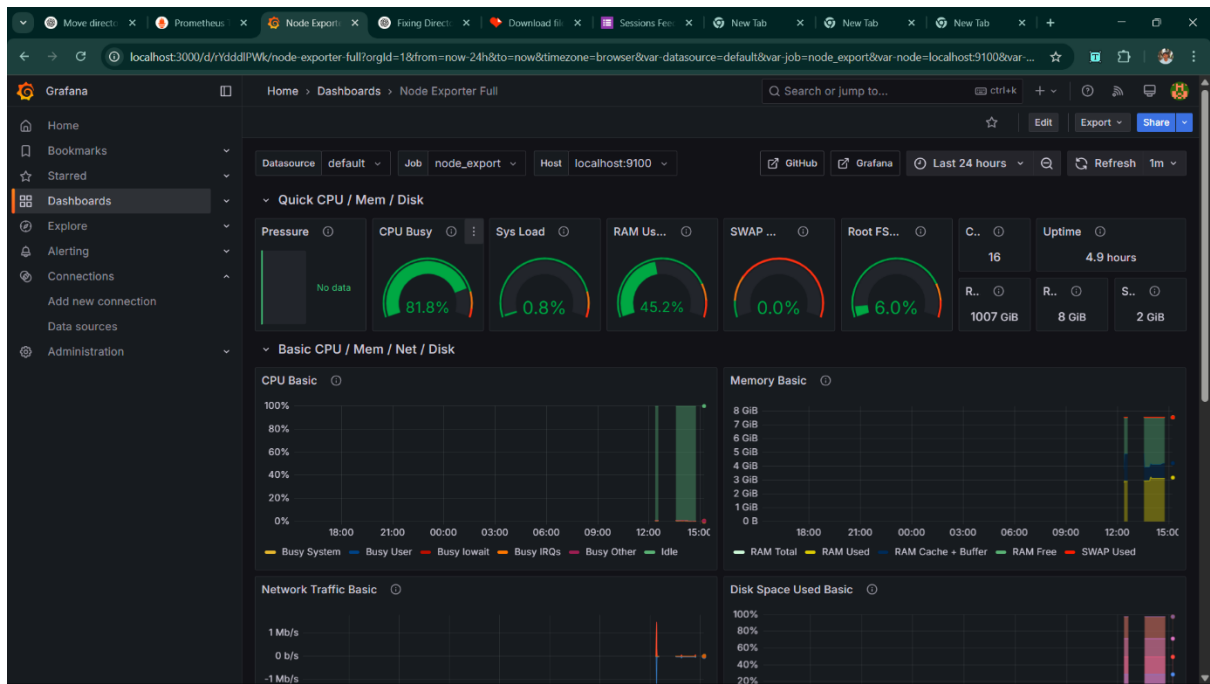
`sudo systemctl start grafana-server`

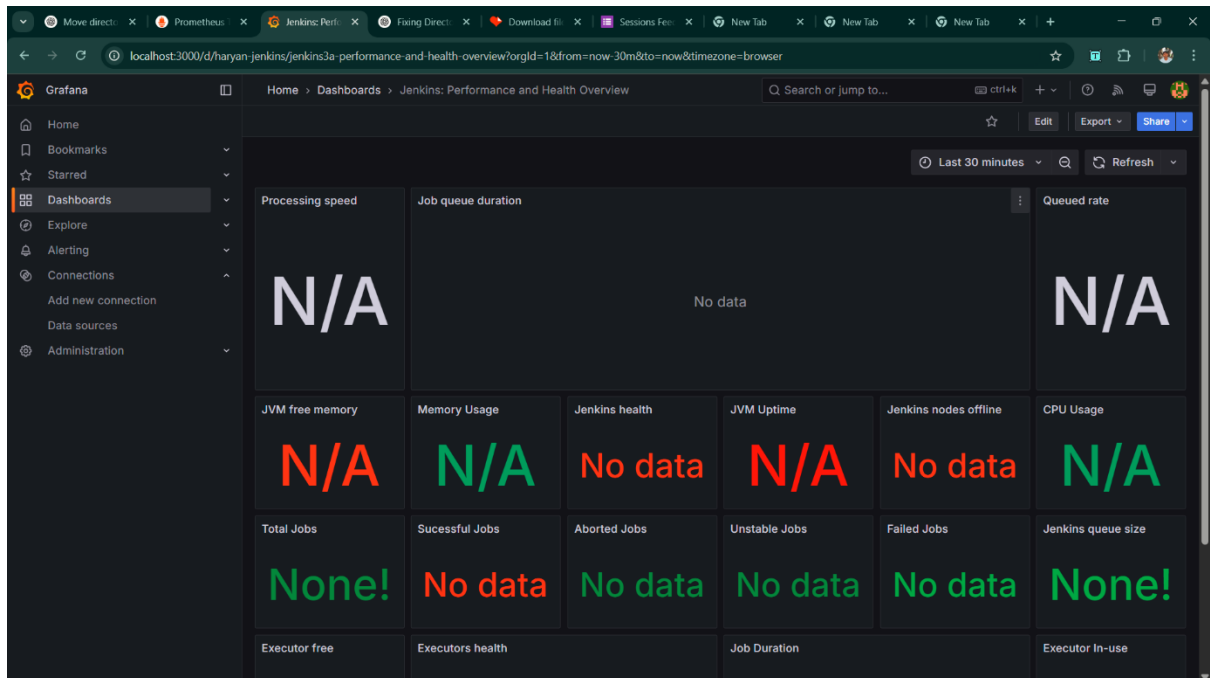
`sudo systemctl status grafana-server`

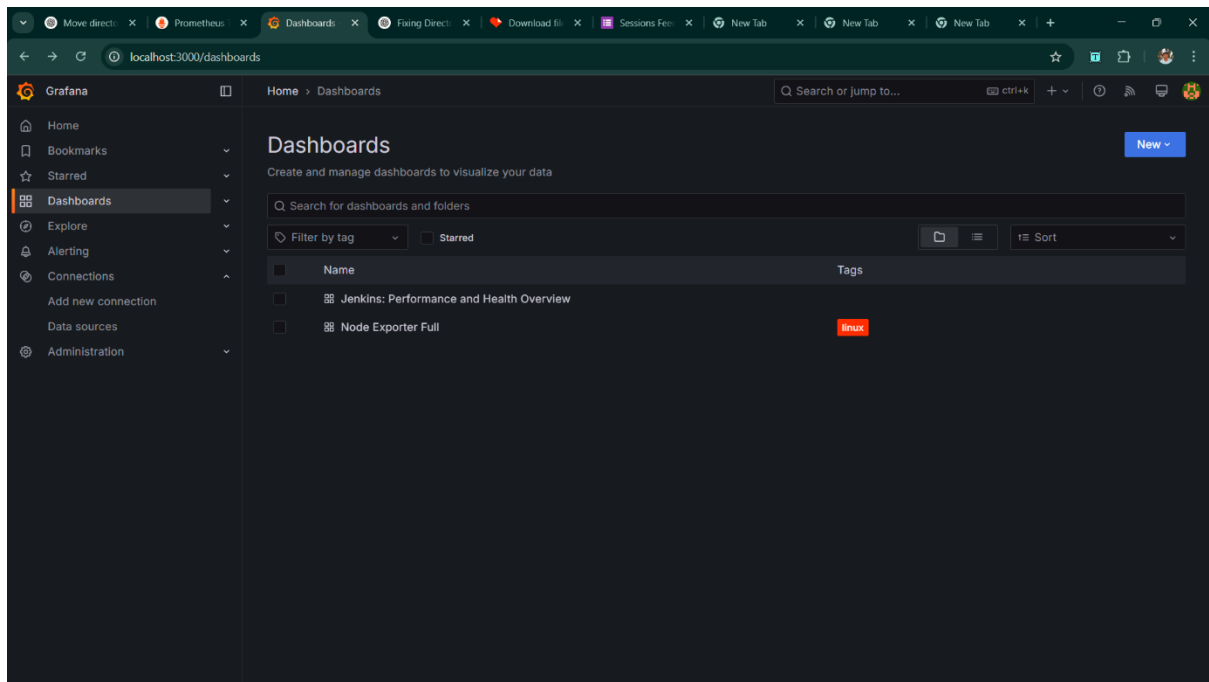
```
ubuntu@prometheus-2.47.1-linux-amd64:~$ sudo systemctl status grafana-server
grafana-server.service - Grafana instance
Loaded: loaded (/lib/systemd/system/grafana-server.service; enabled; vendor preset: enabled)
Active: active (running) since Sat 2025-03-22 12:23:24 IST; 1h 13min ago
Docs: http://docs.grafana.org
Main PID: 51081 (grafana)
Tasks: 17 (limit: 9299)
Memory: 77.8M
CGroup: /system.slice/grafana-server.service
└─51081 /usr/share/grafana/bin/grafana server --config=/etc/grafana/grafana.ini

Mar 22 12:23:32 NIDHARSAN grafana[51081]: logger=installer.fs t=2025-03-22T13:35:16.135Z
Mar 22 12:23:32 NIDHARSAN grafana[51081]: logger=plugins.registration t=2025-03-22T13:35:16.135Z
Mar 22 12:23:32 NIDHARSAN grafana[51081]: logger=plugin.backgroundinstaller t=2025-03-22T13:35:16.135Z
Mar 22 12:23:32 NIDHARSAN grafana[51081]: logger=plugin.backgroundinstaller t=2025-03-22T13:35:16.135Z
Mar 22 12:23:33 NIDHARSAN grafana[51081]: logger=plugin.installer t=2025-03-22T13:35:16.135Z
Mar 22 12:23:33 NIDHARSAN grafana[51081]: logger=installer.fs t=2025-03-22T13:35:16.135Z
Mar 22 12:23:33 NIDHARSAN grafana[51081]: logger=plugins.registration t=2025-03-22T13:35:16.135Z
Mar 22 12:23:33 NIDHARSAN grafana[51081]: logger=plugin.backgroundinstaller t=2025-03-22T13:35:16.135Z
Mar 22 12:24:56 NIDHARSAN grafana[51081]: logger=infra.usagestats t=2025-03-22T13:35:16.135Z
Mar 22 13:35:16 NIDHARSAN grafana[51081]: logger=cleanup t=2025-03-22T13:35:16.135Z
```



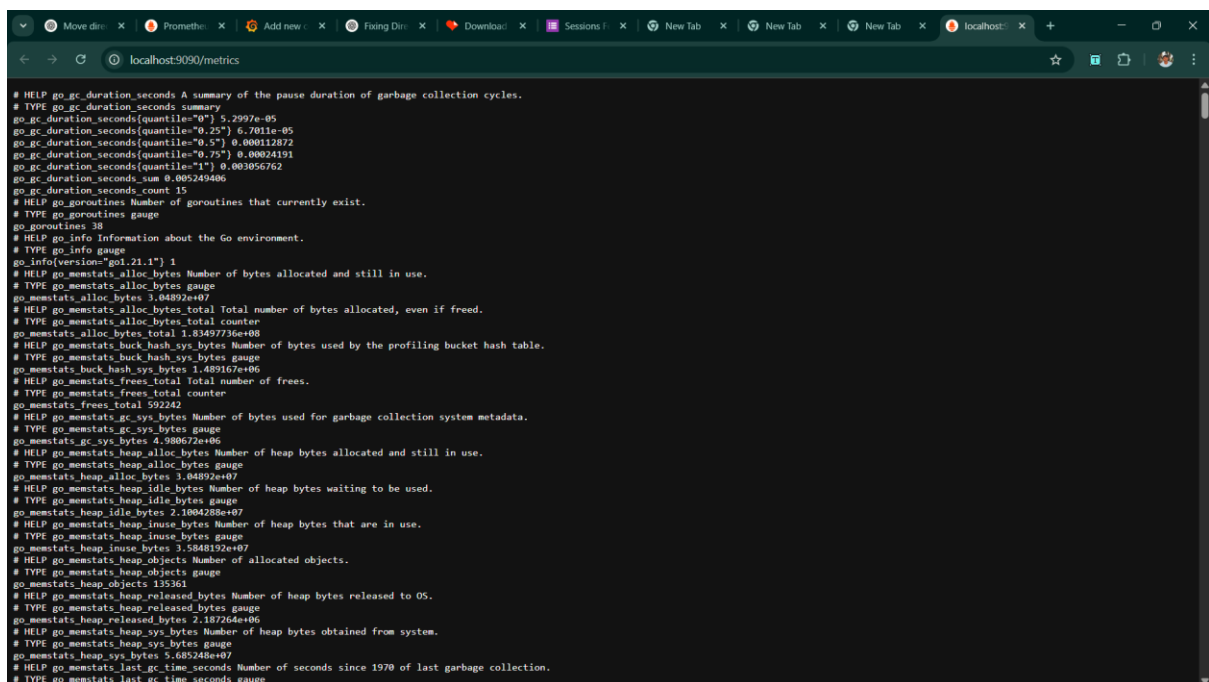




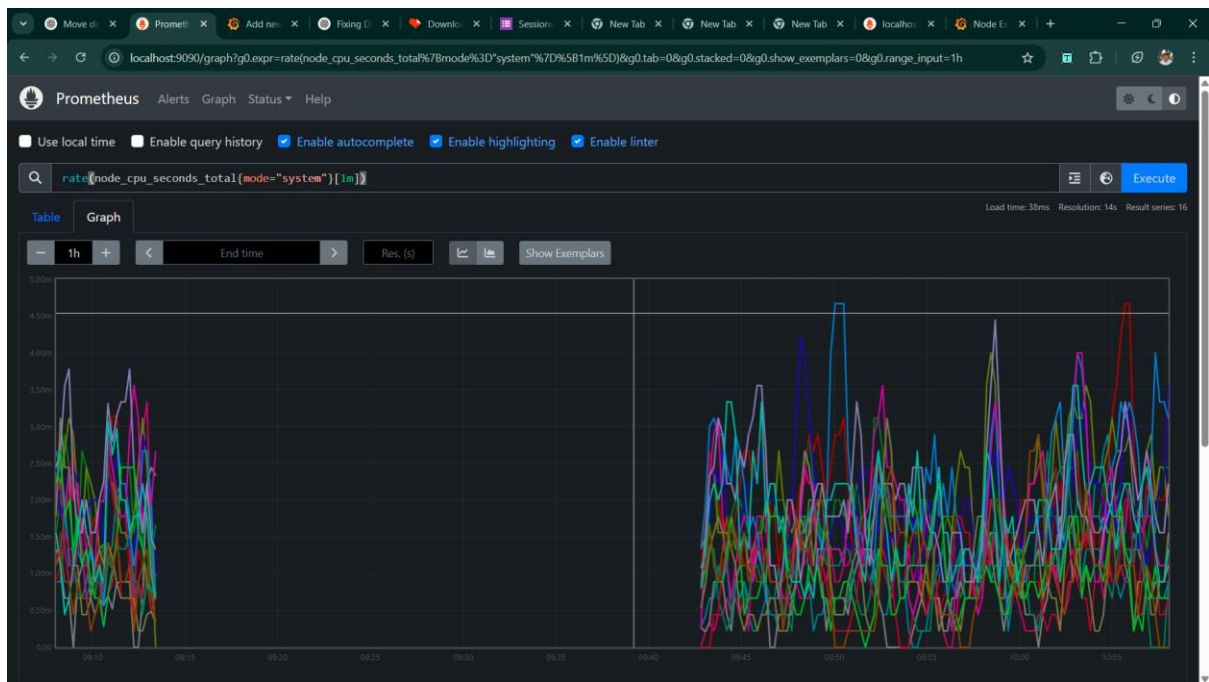


Metrics Running and Verifying:

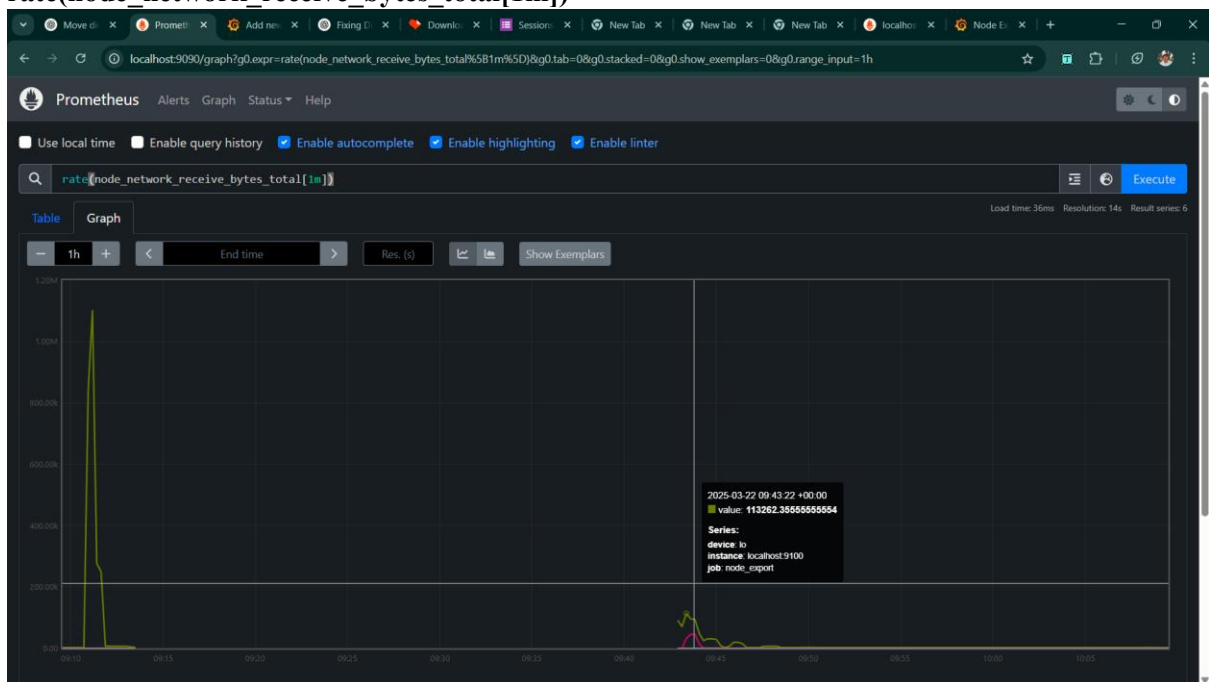
- <http://localhost:9090/metrics>



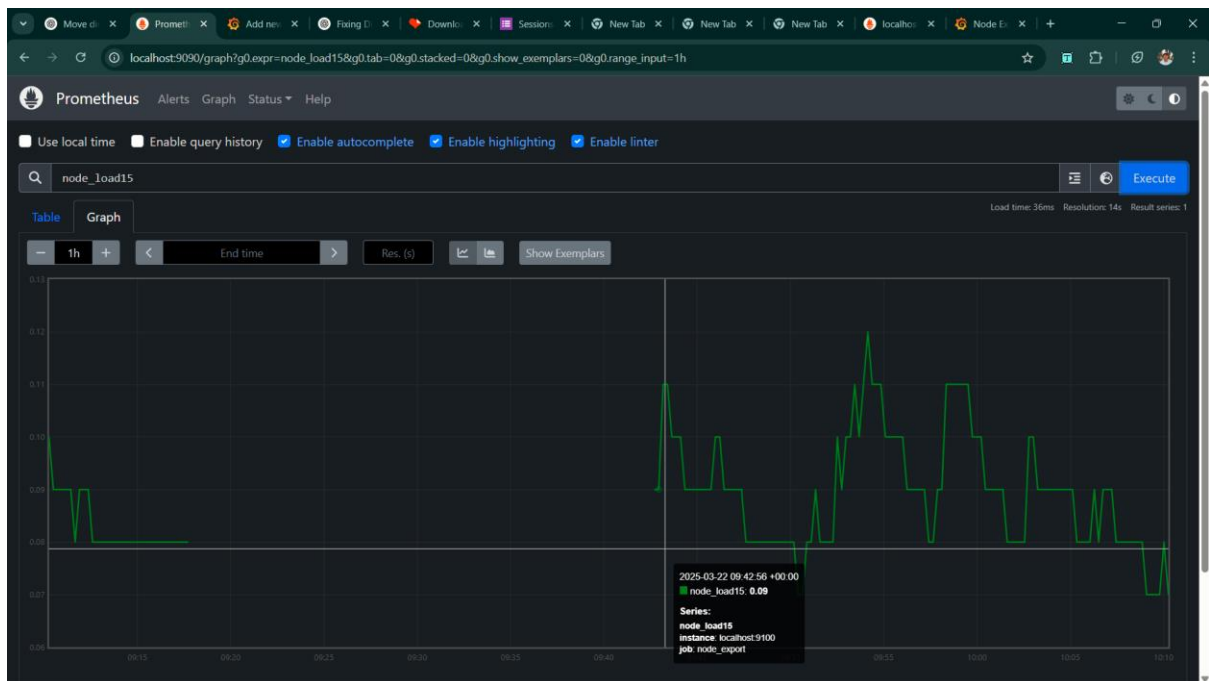
```
rate(node_cpu_seconds_total{mode="system"}[1m])
```



```
rate(node_network_receive_bytes_total[1m])
```

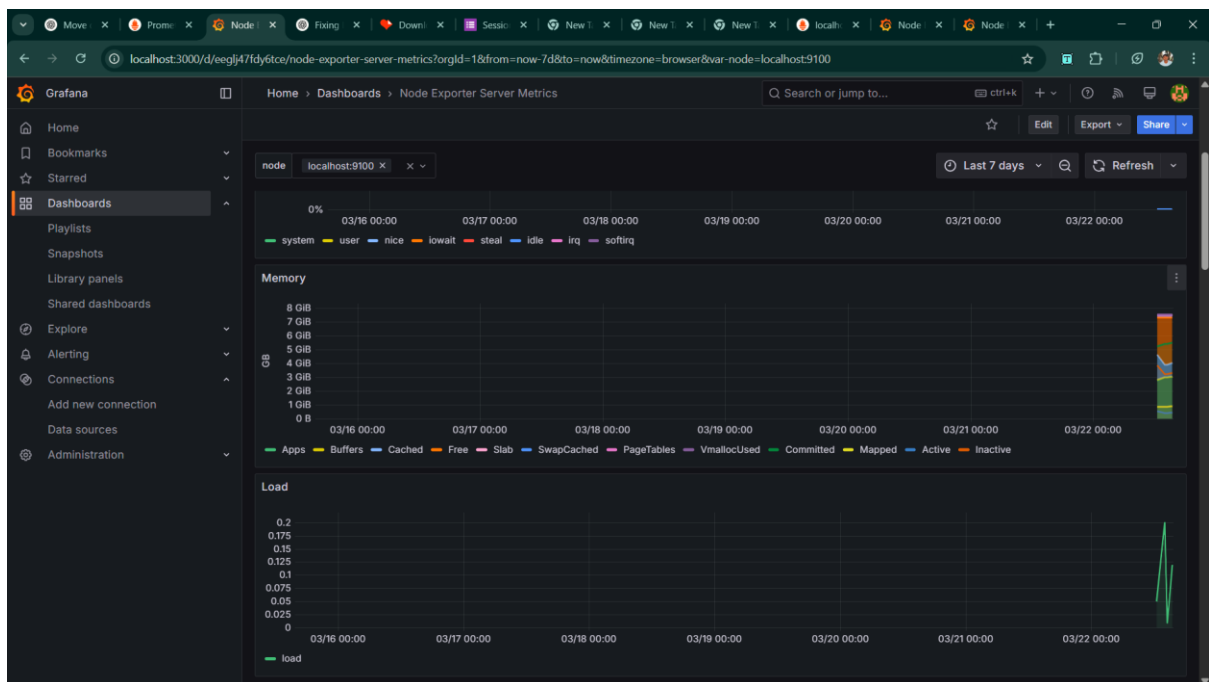


node_load15

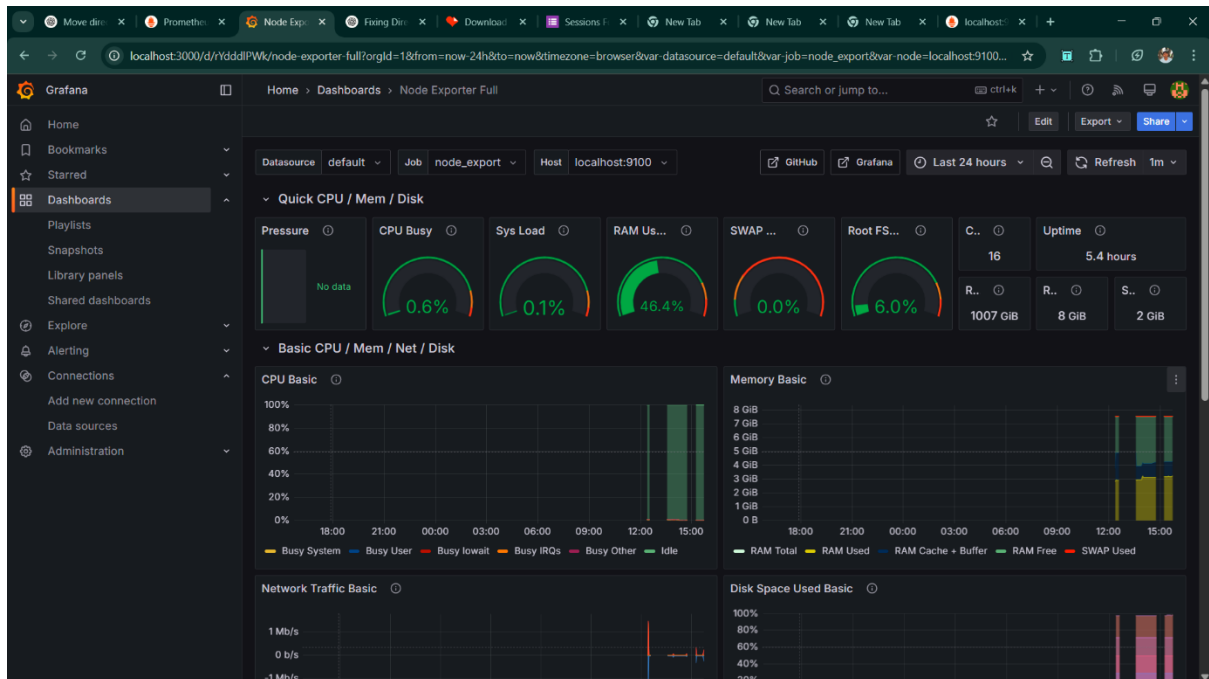


Dashboards:

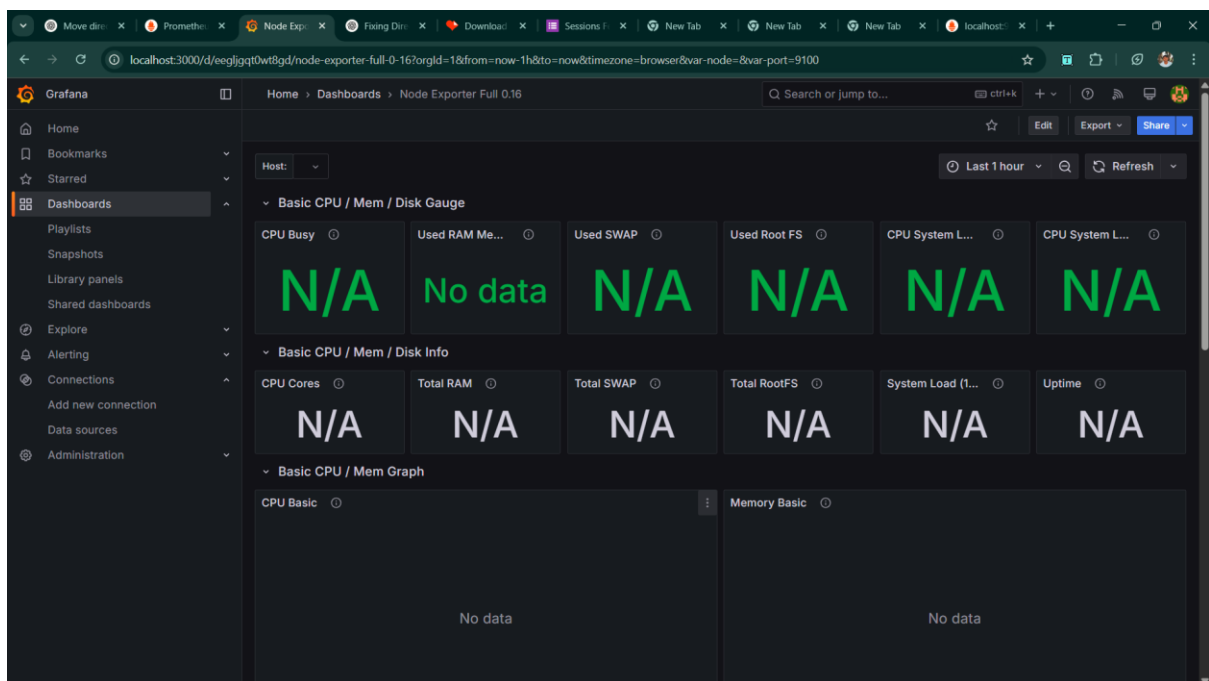
<https://grafana.com/dashboards/405>



<https://grafana.com/dashboards/1860>



<https://grafana.com/dashboards/5174>



<https://grafana.com/dashboards/9096>

